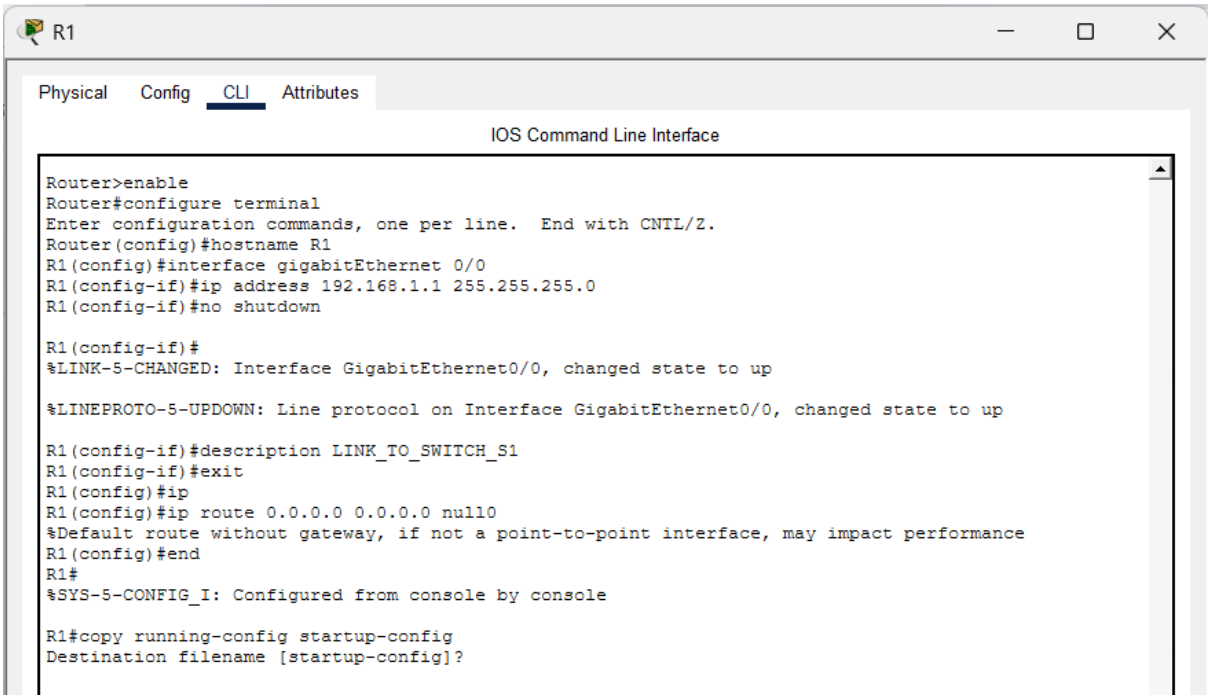
LAB 1 REPORT *Last Updated: 16 Dec 2025*

▪LAB INSTRUCTION STEPS (บอส ปวริศช์ ประมวล 673380278-9)

PART 1: DEVICE CONFIGURATION

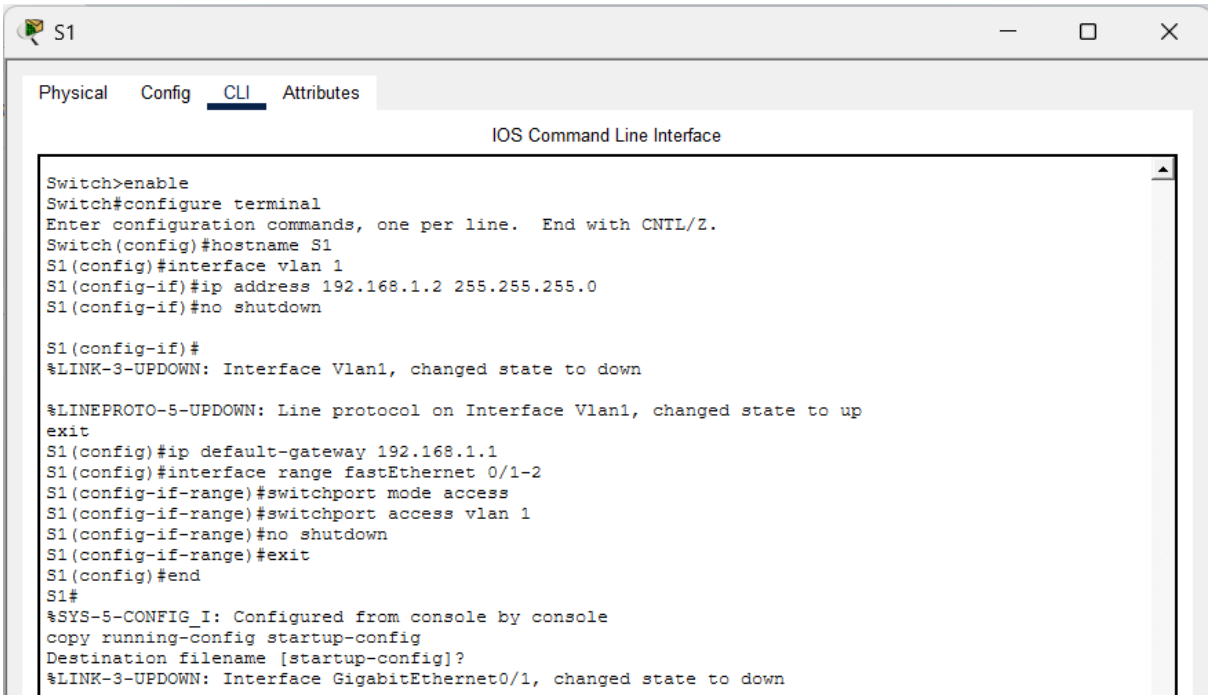
Step 1: Router Configuration

```
None
Router> enable
Router# configure terminal
Router(config)# hostname R1
R1(config)# interface gigabitEthernet 0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
R1(config-if)# description LINK_TO_SWITCH_S1
R1(config-if)# exit
R1(config)# ip route 0.0.0.0 0.0.0.0 null0  ! Static default route for
completeness
R1(config)# end
R1# copy running-config startup-config
```



Step 2: Switch Configuration

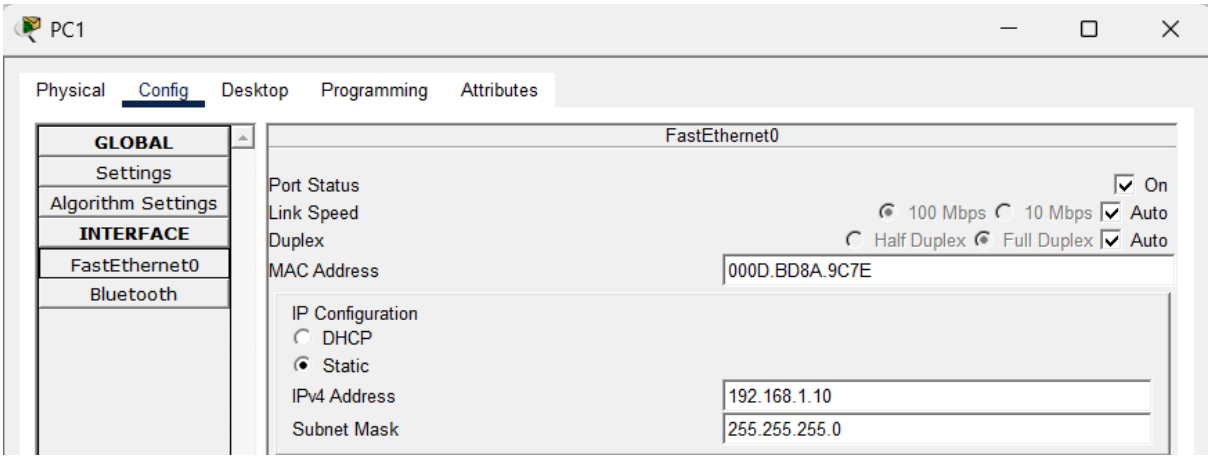
```
None
Switch> enable
Switch# configure terminal
Switch(config)# hostname S1
S1(config)# interface vlan 1
S1(config-if)# ip address 192.168.1.2 255.255.255.0
S1(config-if)# no shutdown
S1(config-if)# exit
S1(config)# ip default-gateway 192.168.1.1
S1(config)# interface range fastEthernet 0/1-2
S1(config-if-range)# switchport mode access
S1(config-if-range)# switchport access vlan 1
S1(config-if-range)# no shutdown
S1(config-if-range)# exit
S1(config)# end
S1# copy running-config startup-config
```



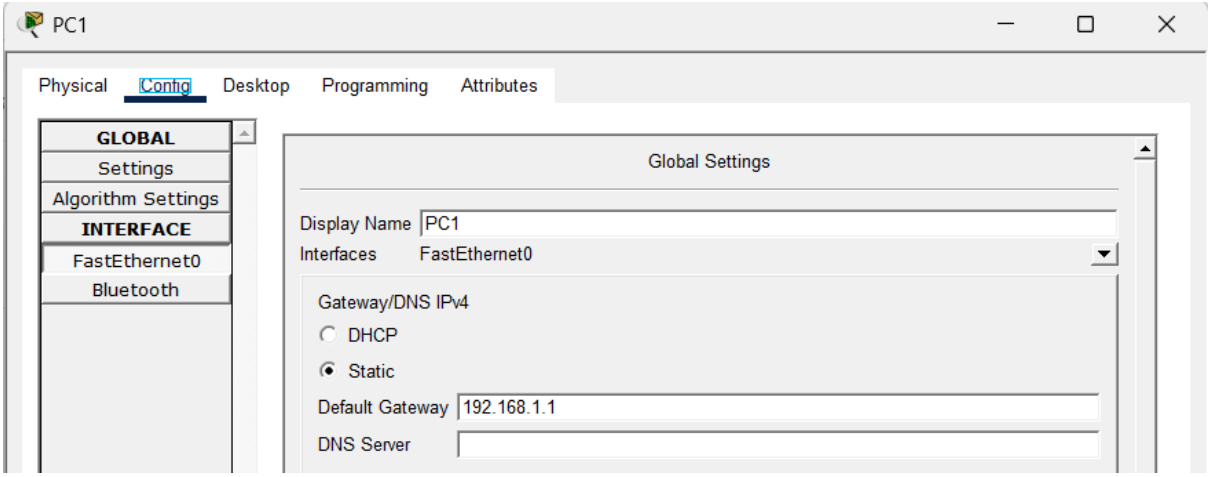
Step 3: PC Configuration

PC1:

- IP Address: 192.168.1.10
- Subnet Mask: 255.255.255.0

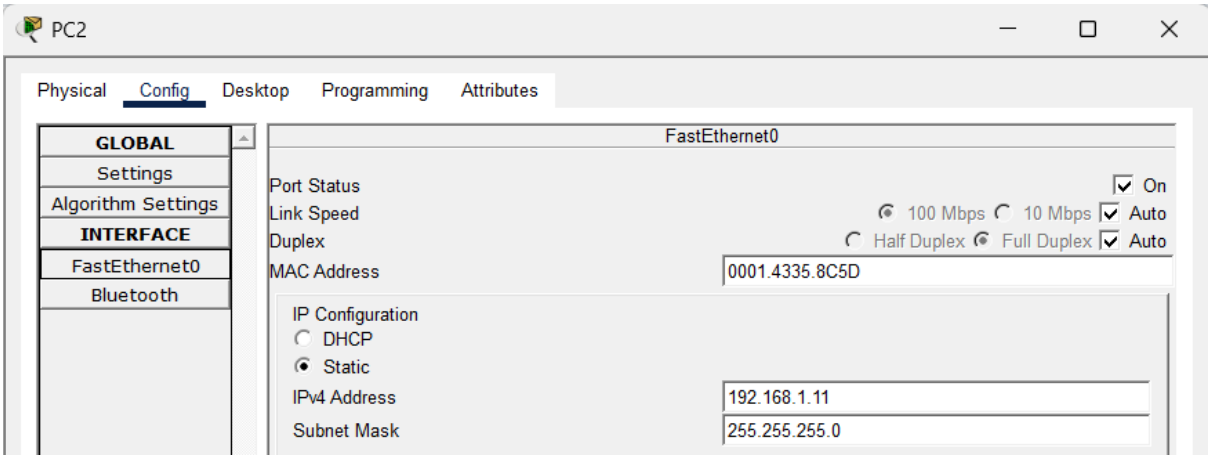


- Default Gateway: 192.168.1.1

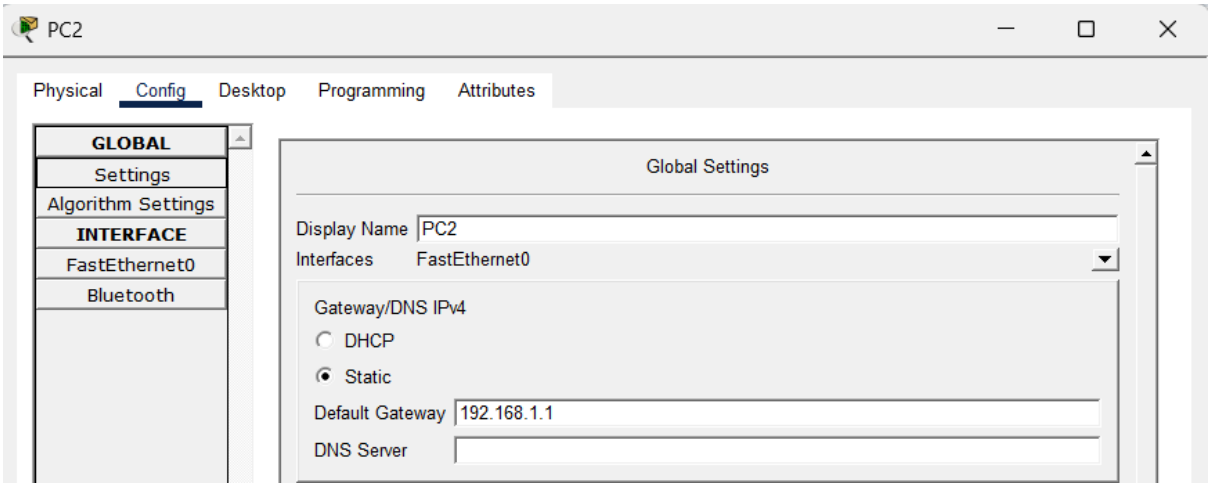


PC2:

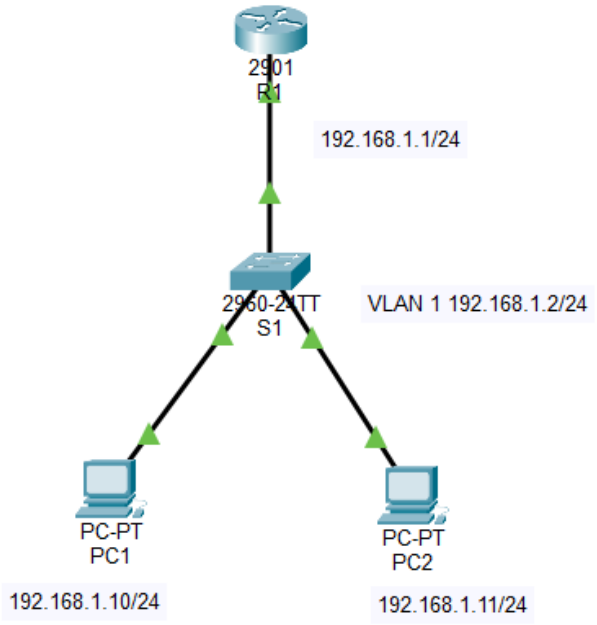
- IP Address: 192.168.1.11
- Subnet Mask: 255.255.255.0



- Default Gateway: 192.168.1.1



Final topology :



PART 2: CONNECTIVITY VERIFICATION

Step 4: Basic Connectivity Tests

```
None
! From PC1 Command Prompt:
PC1> ping 192.168.1.11  ! Verify PC1 to PC2 connectivity
PC1> ping 192.168.1.1   ! Verify PC1 to R1 connectivity
PC1> ping 192.168.1.2   ! Verify PC1 to S1 connectivity

! From R1:
R1# show ip interface brief
! Verify G0/0 status: Protocol UP, Line Protocol UP
! IP address correctly assigned

! From S1:
S1# show vlan brief
! Verify VLAN 1 exists and ports Fa0/1-2 are assigned
S1# show mac address-table
! Observe MAC addresses learned on Fa0/1 and Fa0/2
```

▪LAB DELIVERABLES

Worksheet 1: Configuration Verification (673380293-3 / โอห์ม / ศิระพัทธ์ วงศ์วิวัฒน์เสรี)

Checkpoint	Command Used	Expected Result	Actual Result	Pass/Fail
R1 Interface Status	`show ip int brief`	G0/0 UP/UP	G0/0 Up/UP	Pass
S1 VLAN Config	`show vlan brief`	VLAN 1 Active	VLAN 1 Active	Pass
PC1 Connectivity	`ping 192.168.1.11`	Success (4 replies)	Success (4 received)	Pass
ARP Resolution	`arp -a` on PC1	2+ entries	3 entries	Pass
MAC Learning	`show mac address-table` on S1	2 dynamic entries	VLAN 1 - 3 dynamic entries	Pass

R1 Interface Status :

```
R1>show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     192.168.1.1     YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
```

S1 VLAN Config :

```
S1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

PC1 Connectivity :

```
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

ARP Resolution :

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.1           0030.a3c2.6401        dynamic
  192.168.1.2           00d0.58a5.eec7        dynamic
  192.168.1.11          0060.3e39.8b04        dynamic
```

MAC Learning :

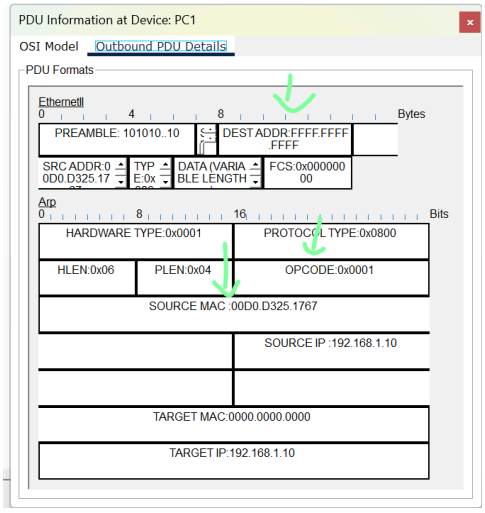
```
S1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0030.a3c2.6401    DYNAMIC     Fa0/3
   1    0060.3e39.8b04    DYNAMIC     Fa0/2
   1    00d0.d325.1767    DYNAMIC     Fa0/1
```

Worksheet 2: Protocol Analysis (673380293-3 / โอห์ม / ศิระพัทธ์ วงศ์วิวัฒน์เสรี)

Capture this data during simulation:

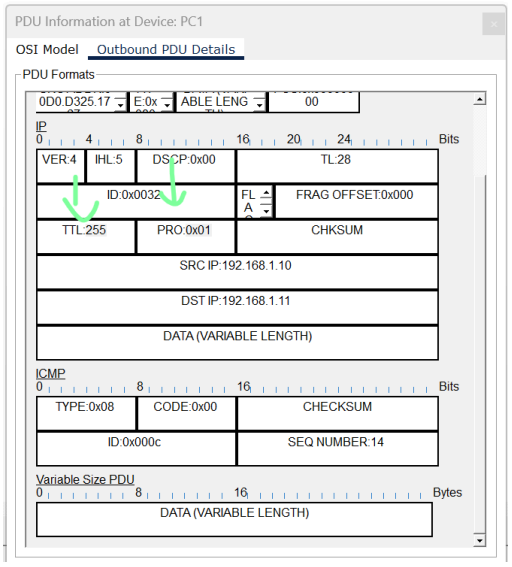
1. **ARP Request Frame:**

   - Source MAC: 00D0.D325.1767
   - Destination MAC: FF:FF:FF:FF:FF:FF
   - ARP Opcode: 0x0001



2. **ICMP Echo Packet:**

   - Time to Live (TTL): 255
   - Protocol Field: 0x01 (Value = 1)



3. **Encapsulation Chain:** Data →ICMP Header → IP Header → Ethernet Header → Bits

Worksheet 3: Troubleshooting Journal

Problem Injected	Symptoms Observed	Diagnostic Commands Used	Root Cause	Solution
Wrong Subnet Mask	Cannot ping other devices on the network	`ping`, `arp -a`	Incorrect subnet mask	Correct the subnet mask
Interface Shutdown	No network connectivity, Ping fails immediately	`show ip int brief`	Router or switch interface is down	Enable the interface using “no shutdown”
Incorrect Gateway	Can ping local devices but cannot reach outside the subnet	`ipconfig`, `tracert`	PC configured with an incorrect default gateway IP	Set the default gateway to 192.168.1.1 (R1)

▪VERIFICATION COMMANDS CHECKLIST (673380515-1 / อ๋อง / ศรัณย์ พาพรชัย)

Router (R1):

☑ show running-config

```
interface GigabitEthernet0/0/0
 description LINK_TO_SWTICH_S1
 no ip address
 duplex auto
 speed auto
!
interface GigabitEthernet0/0/1
 ip address 192.168.1.1 255.255.255.0
 duplex auto
 speed auto
!
interface GigabitEthernet0/0/2
 no ip address
 duplex auto
 speed auto
 shutdown
!
interface Vlan1
 no ip address
 shutdown
!
router rip
!
ip classless
ip route 0.0.0.0 0.0.0.0 Null0
!
ip flow-export version 9
```

Interface shown GigabitEthernet0/0/1 connecting to
ip address 192.168.1.1
Submask 255.255.255.0

☑ show ip interface brief

```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0/0   unassigned      YES manual down                  down
GigabitEthernet0/0/1   192.168.1.1     YES manual up                    up
GigabitEthernet0/0/2   unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1#
```

Ip interface shown GigabitEthernet0/0/1 status is up

☑ show arp

```
R1#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.1.1             -   0001.64E9.D202  ARPA   GigabitEthernet0/0/1
R1#
```

The Arp (Address Resolution Protocol) show the Hardware Address (0001.64E9.D202)

- ☑ show ip route

```
R1#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.1.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.1.1/32 is directly connected, GigabitEthernet0/0/1
S*   0.0.0.0/0 is directly connected, Null0

R1#
```

Router IP : 192.168.1.0/24
Connected : 192.168.1.1/32

Switch (S1):

- ☑ show running-config

```
S1#show running-config
Building configuration...

Current configuration : 1170 bytes
!
version 15.0
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname S1
!
!
!
```

```
interface GigabitEthernet0/2
!
interface Vlan1
 ip address 192.168.1.2 255.255.255.0
!
ip default-gateway 192.168.1.1
!
!
!
!
line con 0
!
line vty 0 4
 login
line vty 5 15
 login
!
!
!
!
end
```

Hostname : S1
Interface GigabitEthernet0/2 connected
Vlan ip address : 192.168.1.2
Subnet mask : 255.255.255.0
Default gateway : 192.168.1.1

- ☑ show vlan brief

```
S1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
S1#
```

The command shows the ports that activate and status.

- ☑ show mac address-table

```
S1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0001.64e9.d202    DYNAMIC     Gig0/1
S1#
```

MAC address : 0001.64e9.d202
Type : Dynamic
Connected port : GigabitEthernet0/1

- ☑ `show interfaces status`

```
S1#show interfaces status
Port      Name               Status       Vlan       Duplex  Speed Type
Fa0/1                        connected    1          a-full  a-100 10/100BaseTX
Fa0/2                        connected    1          a-full  a-100 10/100BaseTX
Fa0/3                        notconnect   1          auto    auto  10/100BaseTX
Fa0/4                        notconnect   1          auto    auto  10/100BaseTX
Fa0/5                        notconnect   1          auto    auto  10/100BaseTX
Fa0/6                        notconnect   1          auto    auto  10/100BaseTX
Fa0/7                        notconnect   1          auto    auto  10/100BaseTX
Fa0/8                        notconnect   1          auto    auto  10/100BaseTX
Fa0/9                        notconnect   1          auto    auto  10/100BaseTX
Fa0/10                       notconnect   1          auto    auto  10/100BaseTX
Fa0/11                       notconnect   1          auto    auto  10/100BaseTX
Fa0/12                       notconnect   1          auto    auto  10/100BaseTX
Fa0/13                       notconnect   1          auto    auto  10/100BaseTX
Fa0/14                       notconnect   1          auto    auto  10/100BaseTX
Fa0/15                       notconnect   1          auto    auto  10/100BaseTX
Fa0/16                       notconnect   1          auto    auto  10/100BaseTX
Fa0/17                       notconnect   1          auto    auto  10/100BaseTX
Fa0/18                       notconnect   1          auto    auto  10/100BaseTX
Fa0/19                       notconnect   1          auto    auto  10/100BaseTX
Fa0/20                       notconnect   1          auto    auto  10/100BaseTX
Fa0/21                       notconnect   1          auto    auto  10/100BaseTX
Fa0/22                       notconnect   1          auto    auto  10/100BaseTX
Fa0/23                       notconnect   1          auto    auto  10/100BaseTX
Fa0/24                       notconnect   1          auto    auto  10/100BaseTX
Gig0/1                       connected    1          a-full  a-100 10/100/1000BaseTX
Gig0/2                       notconnect   1          auto    auto  10/100/1000BaseTX

S1#
```

Interface shows the status of entire port in S1
F1 connected with a-full and speed a-100

End Devices:

- ☑ `ipconfig /all` (or equivalent)

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ipconfig /all

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Physical Address................: 00E0.A379.935A
   Link-local IPv6 Address.........: FE80::2E0:A3FF:FE79:935A
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.1.10
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.1.1
   DHCP Servers....................: 0.0.0.0
   DHCPv6 IAID.....................:
   DHCPv6 Client DUID..............: 00-01-00-01-30-2B-C7-7A-00-E0-A3-79-93-5A
   DNS Servers.....................: ::
                                     0.0.0.0

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Physical Address................: 0001.C903.DDC8
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
   DHCP Servers....................: 0.0.0.0
   DHCPv6 IAID.....................:
   DHCPv6 Client DUID..............: 00-01-00-01-30-2B-C7-7A-00-E0-A3-79-93-5A
   DNS Servers.....................: ::
                                     0.0.0.0


C:\>
```

Ip config command shows the state of PC such as
IP address : 192.168.1.10
Subnet Mask : 255.255.255
Default Gateway : 192.168.1.1

- ☑ `arp -a`

```
C:\> arp -a
No ARP Entries Found
C:\>
```

`No ARP Entries Found : No cached found in PC`

- ☑ `ping` to all devices

```
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

PC1 ping to -> PC2 : PC1 can received 4 packets from PC2 (ping successful)

- ☑ `tracert` (if multi-hop)

```
C:\>tracert 192.168.1.11

Tracing route to 192.168.1.11 over a maximum of 30 hops:

  1   0 ms      0 ms      0 ms      192.168.1.11

Trace complete.

C:\>
```

1 0ms 0ms 0ms 192.168.1.11
This command show the packet sending to destination just only 1 hop

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▪ASSESSMENT QUESTIONS (673380272-1 / โชกุน / ธนภูมิ จันทรา)

1. When PC1 pings PC2, which protocol resolves the MAC address before ICMP operates?
    - **ARP (Address Resolution Protocol)**, it maps the destination IP address to a MAC address so the Ethernet frame can be sent

2. What would happen if you changed the subnet mask on PC1 to 255.255.255.128?
    - PC1 would fail to communicate with PC2

3. Why does the switch not need an IP address for basic frame forwarding?
    - Because a switch forwards traffic using MAC addresses, not IP addresses

4. Which OSI layers are involved when R1 forwards a packet between interfaces?
    - Layers 1, 2, and 3

5. What is the purpose of the TTL field in the IP header you observed?
    - To prevent packets from looping endlessly in the network